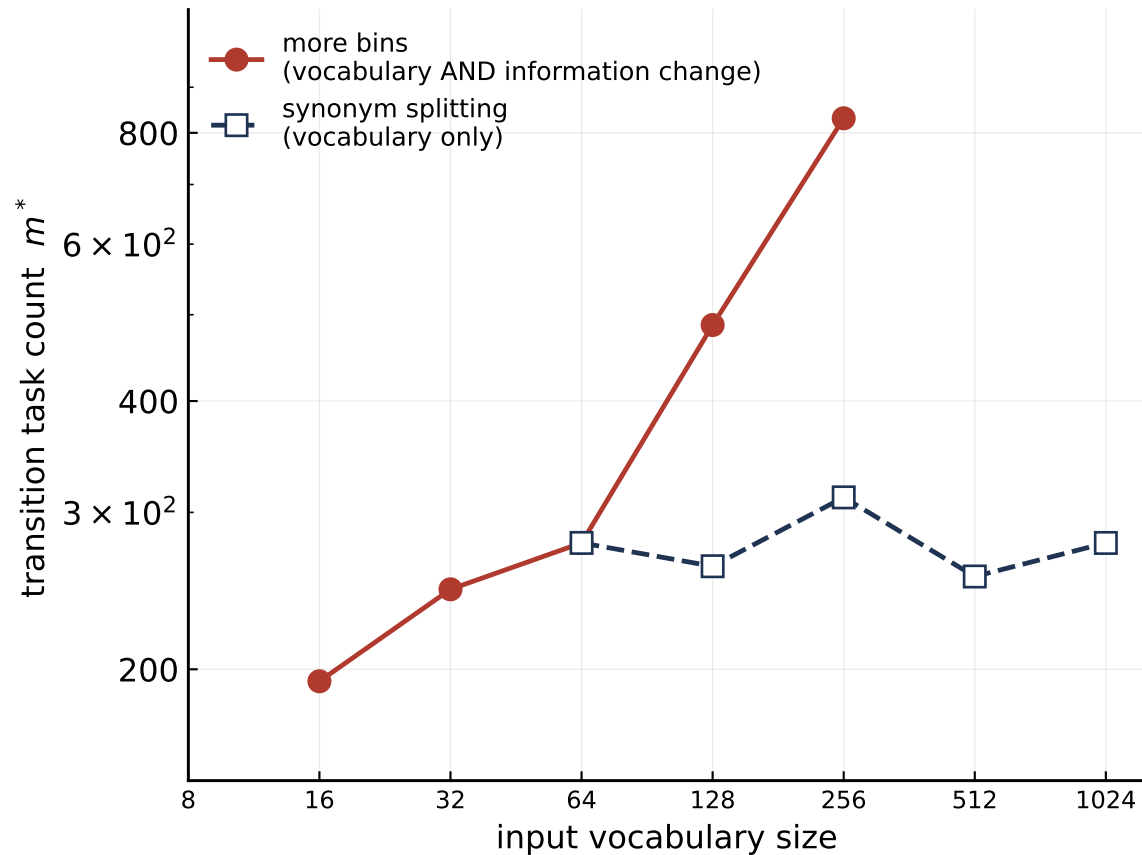
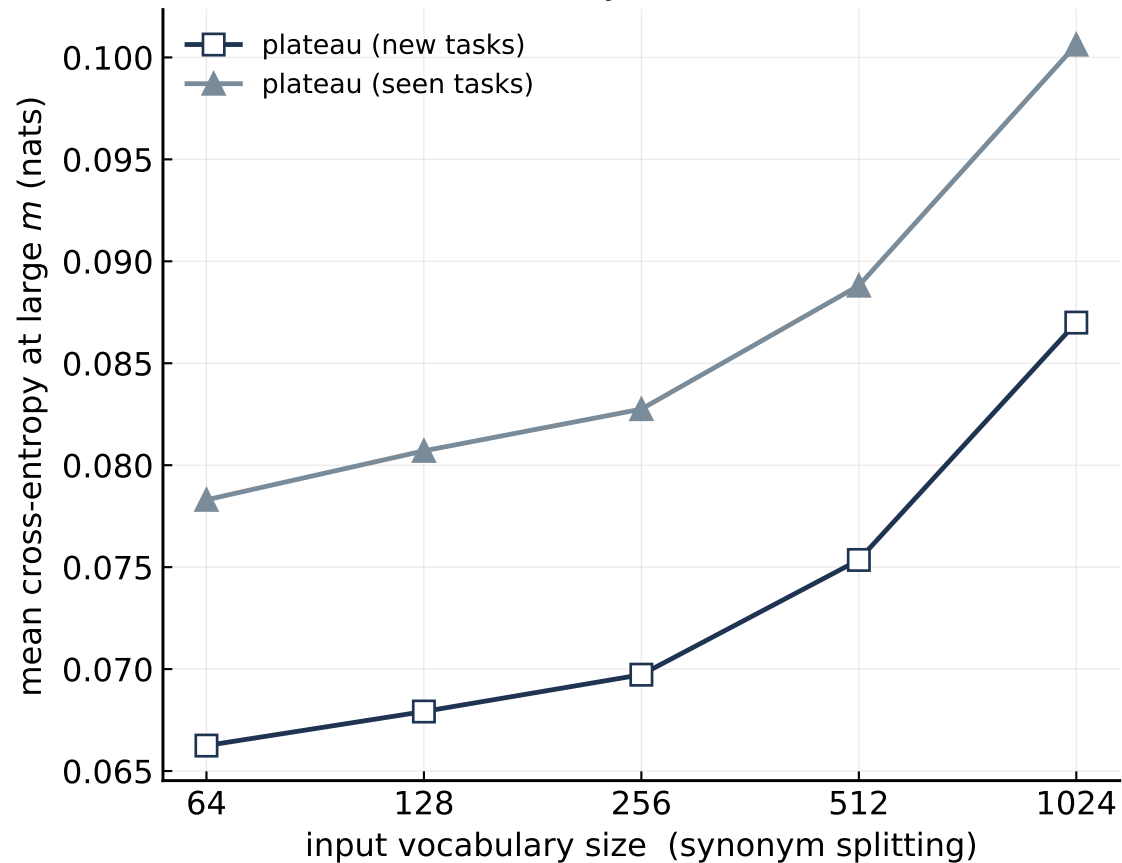


Where the transition sits



What vocabulary inflation does cost



Splitting each of 64 bins into  $j$  interchangeable tokens grows the vocabulary  $j$ -fold while leaving the information, the sequence length and the 64-bin output unchanged. The transition does not move; only the achievable floor drifts. All arms at 320k steps.